

II-JU – Cube27 Endotrivial Block Equivalence/Picard Action and Stable AR Shadow, Gauss Pressure/Khinchin State, Codimension-3 Branch Transitions, Golden Comparison Cochains, PLR–Pisano $3C_4 \rightarrow 1 + 2$ Discriminator, Direct 2-Adic Gauss Sum, and Payload Search

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Abstract

II-JU freezes II-JT and preserves the primary carrier E_{13} cuboctahedral/FCC $\leftrightarrow$ Cube27 $\leftrightarrow$ O_{14} resolved stella-octangula. The complete endotrivial group $C_4 \times C_2$ is made into an explicit Picard action shadow: two Ω -cycles of length four are exchanged by the total-sign twist and split into four Auslander–Reiten $\tau = \Omega^2$ pairs. The Gauss directive carrier gains its pressure tangent and a certified Khinchin-state enclosure. Epstein–Vogt branch transitions are extracted from the source polynomial and the exact half-turn flow involution. Golden–Hodge extension gains satisfy the new exact formula $2^{\text{rank}(t-I)} - 1$, localizing the native/small comparison obstruction. The user’s proposed PLR discrimination is resolved: the three regular C_4 choices are canonically indexed by the three nonidentity elements of the unique regular V_4 ; the Fibonacci mod-3 antiperiod from the pinned recurrence repository supplies an external diagonal sign $11 \in U(12)$, and after a declared $V_4^{PLR} \rightarrow U(12)$ frame it splits $3C_4 = C_{11} \sqcup \{C_5, C_7\}$. This is conditional, not a native PLR selector. Finally, the previously forced 2-adic sign is independently replayed by a two-unit local Gauss sum, giving $\epsilon_2 = -1$ for $k = 1, 3$ in the frozen different-normalized convention. Stone/Fibonacci still provide no ABI-complete project packet.

1 Frozen entry and typing discipline

II-JT is frozen at

$$\boxed{974 \text{ PASS, } 84 \text{ ZERO, } 288 \text{ SEPARATED, } 256 \text{ REFUTED, } 38 \text{ NT}}$$

$$\boxed{492 \text{ PROVEN-MERGEABLE, } 315 \text{ PROVEN-NOT-MERGEABLE}}.$$

Exactly 37 project-value gates remain active. The primary geometric ordering stays

$$\boxed{E_{13}^{\text{cubo/FCC}} \longleftrightarrow \text{Cube27} \longleftrightarrow O_{14}^{\text{resolved stella}}.}$$

All PLR, ternary, Golden–Hodge, Pisano and local-number-field comparisons below are explicit adapters, shadows or obstructions. Matching cardinalities alone never identify carriers.

2 Cube27: Picard action, block equivalence, and AR shadow

II-JT closed

$$T_{\mathbb{F}_3}(B_3) \cong \text{Pic}(\text{stmod } \mathbb{F}_3[B_3]) \cong C_4 \times C_2.$$

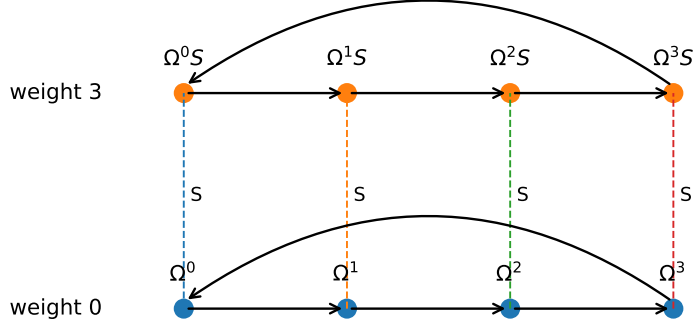
Write its eight endotrivial classes as $(i, j) \in \mathbb{Z}/4 \times \mathbb{Z}/2$, where Ω increments i and the total-sign line S increments j . Then

$$\Omega(i, j) = (i + 1, j), \quad S(i, j) = (i, j + 1), \quad \tau = \Omega^2.$$

Hence the full endotrivial shadow is exactly two oriented 4-cycles exchanged by S , while τ has four orbits of size 2:

$$\boxed{2C_4^\Omega \xleftrightarrow{S} \text{sign-twisted block, } 4C_2^\tau.}$$

Tensoring with S is a stable block equivalence between the two full-3-defect H-character blocks of weights 0 and 3. The action is replayed on all eight classes.



Guard. This is the complete Picard/AR *shadow on the eight endotrivial classes*. It is not the full stable Auslander–Reiten quiver of every finite-dimensional $\mathbb{F}_3[B_3]$ -module.

3 Gauss-tail pressure and the Khinchin directive state

On the previously frozen directive-slope carrier, the invariant state is

$$d\mu(x) = \frac{dx}{\log 2 (1+x)}.$$

For the geometric potential $-s \log |G'|$, the pressure tangent at $s = 1$ is

$$P(1) = 0, \quad P'(1) = - \int \log |G'| d\mu = - \frac{\pi^2}{6 \log 2}.$$

The Khinchin cocycle is $\kappa(x) = \log a_1(x)$, with

$$\mu(a_1 = n) = \log_2 \frac{(n+1)^2}{n(n+2)}.$$

A 400 000-digit prefix plus a monotone analytic tail enclosure gives

$$2.6854520009091 < K < 2.6854520012336,$$

where $K = \exp \int \kappa d\mu$. Thus the directive state has one frozen Gauss-ergodic component and a reproducible multiplicative digit scale $K \approx 2.6854520011$.

4 Codimension three: actual branch-transition extraction

For the Epstein–Vogt octagonal base,

$$\psi(x, y) = (4 - x^2)(4 - y^2)(9 - (x + y)^2)(9 - (x - y)^2).$$

Direct symbolic replay proves invariance under

$$(x, y) \mapsto (-x, y), \quad (x, -y), \quad (y, x), \quad (-x, -y),$$

so the support base carries the expected D_4 transition shadow. The source’s eight branch arcs alternate:

$$Q_{1,3,5,7} : S^3 \text{ pieces with slope } \pm 1, \quad Q_{2,4,6,8} : T^2 \text{ pieces from collapsed circles.}$$

More importantly, Epstein–Vogt give the exact flow-compatible involution

$$T(x, y, u_1, u_2, w_1, w_2, z) = (-x, -y, u_2, u_1, w_2, w_1, z),$$

which preserves the defining data and vector field; under cyclic branch indexing it exchanges opposite branches $Q_i \leftrightarrow Q_{i+4}$.

Thus the actual source closes support-level D_4 transitions and the opposite-branch flow transition. A complete atlas of the true non-Hausdorff orbit space, with all adjacent flow transition maps, remains **SEPARATED**.

5 Golden-Hodge: why the multiplicative defect is exactly 1 or 3

The exact native/small multiplicative-extension profile remains

$$\delta = (0, 0, 0, 1, 3, 1, 3).$$

For the four order-32 stabilizers, write $G = V \rtimes \langle t \rangle$ with $V \cong \mathbb{F}_2^4$. The inherited H^1 dimensions are 4, 3, 4, 3. Since

$$\dim H^1(G; \mathbb{F}_2) = 1 + \dim V/(t - I)V = 5 - \text{rank}(t - I),$$

the corresponding ranks are 1, 2, 1, 2. II-JU verifies the stronger formula

$$\delta_G = |\text{im}(t - I) \setminus \{0\}| = 2^{\text{rank}(t-I)} - 1,$$

thereby reproducing

$$1, 3, 1, 3.$$

So gain 1 is the punctured binary line and gain 3 the punctured binary plane.

The required comparison cochain has the exact formal equation

$$d\eta_{a,b} = \smile_{\text{bar}} (\Phi a, \Phi b) - \Phi(a \smile_{\text{small}} b).$$

The right-hand correction ranks are now frozen, and $\eta = 0$ is refuted on all four order-32 carriers. The small model already has

$$H^4 = \langle H^1 H^3, H^2 H^2 \rangle,$$

but a native H^4 coordinate table still needs the residual H^3 bar representatives and actual solutions $\eta_{a,b}$. Hence the comparison-cochain *problem* is now exact, while full native reduction stays **SEPARATED**.

6 PLR/Pisano discriminator: the three C_4 choices really have a square label

On four PLR labels there is one regular V_4 and three regular C_4 subgroups. Every regular C_4 has a unique involution, namely the square of either generator. Exhaustive S_4 replay gives the canonical bijection

$$\{\text{three regular } C_4\} \xrightarrow{C \mapsto C[2]} V_4 \setminus \{1\}.$$

This already upgrades the former cardinality coincidence: the three cyclic choices are canonically indexed by the three nonzero points of the regular binary plane.

The pinned GitHub recurrence analyzer independently reports for Fibonacci mod 3

$$01120221 = 0112 \mid 0221, \quad h = 4,$$

and its test fixture fixes the halves 0112 and 0221. On the companion-state carrier

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad M^4 = -I, \quad M^8 = I.$$

Thus $\langle M^2 \rangle \cong C_4$ and its unique involution is $-I$. Moreover

$$0221 = -0112 \pmod{3},$$

so the two half-words are not independent cyclic carriers; they are the two halves of one antiperiodic orbit. Their ordered residue counts (1, 2, 1) and (1, 1, 2) are exchanged by the mod-3 sign.

Now use the previously certified

$$U(12) = \{1, 5, 7, 11\} \cong V_4, \quad 5 \leftrightarrow \nu_5, \quad 7 \leftrightarrow \nu_7, \quad 11 \leftrightarrow \nu_5 \nu_7.$$

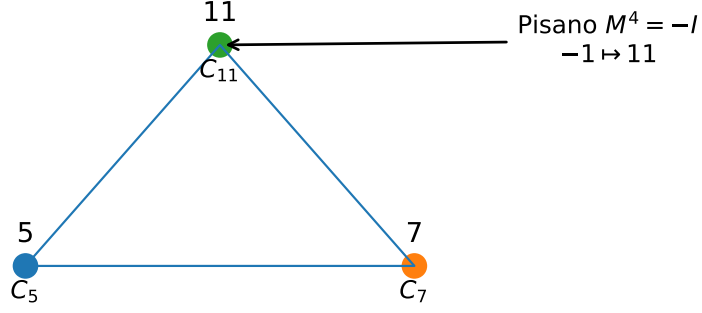
The common sign adapter sends

$$-I \mapsto -1 \equiv 11 \pmod{12}.$$

Therefore, *after choosing* an external group frame $V_4^{PLR} \rightarrow U(12)$, label 11 selects the unique PLR C_4 whose square is the diagonal involution:

$$3C_4 = C_{11} \sqcup \{C_5, C_7\}.$$

This is the requested 1 + 2 discrimination.



after an external $V_4^{PLR} \rightarrow U(12)$ frame: $3C_4 = C_{11} \sqcup \{C_5, C_7\}$

There are exactly six group frames $V_4^{PLR} \rightarrow U(12)$, and across them each of the three PLR C_4 choices is selected by 11 exactly twice. Consequently the split is

PASS conditional on the declared frame, native unframed selector : **SEPARATED**.

The unconditional canonical split is **REFUTED**.

Relation to the Golden defect

The gain-3 carrier is likewise a punctured $\mathbb{F}_2^2$, hence admits a $3 = 1 + 2$ decomposition after choosing one nonzero vector. This is the same abstract binary geometry as the PLR square-index set. But $GL(2, 2) \cong S_3$ gives six possible frames, and no Golden source selects one. Thus the strengthened relation is

same punctured-plane type : **PASS**, canonical cross-frame : **SEPARATED**.

The Pisano label 11 supplies a selector only on the *framed PLR*/ $U(12)$ side; it does not manufacture a Golden cohomology class.

7 Pisano–Theta: direct 2-adic Gauss-sum replay

Let $K = \mathbb{Q}_2(w)$ with $w^2 = -5$ and $\pi = 1 + w$. Then $N(\pi) = 6$, the different exponent is $d = 2$, and the odd ring-class characters $k = 1, 3$ have local conductor exponent $a = 2$. In the frozen local convention the unit quotient has two classes represented by $1, w$ with character values $1, -1$.

The finite Tate sum uses denominator $\pi^{a+d} = \pi^4$. Exact arithmetic gives

$$\begin{aligned} \pi^4 &= -4 - 16w, & \frac{1}{\pi^4} &= -\frac{1}{324} + \frac{w}{81}, & \text{Tr } \frac{1}{\pi^4} &= -\frac{1}{162}, \\ \frac{w}{\pi^4} &= -\frac{5}{81} - \frac{w}{324}, & \text{Tr } \frac{w}{\pi^4} &= -\frac{10}{81}. \end{aligned}$$

For the standard 2-adic additive character transported by trace, these two traces contribute -1 and $+1$. Therefore the normalized two-unit Gauss sum is

$$2^{-1}((-1) - (+1)) = -1.$$

Hence the sign previously forced globally is now independently replayed:

$$\epsilon_2(k) = -1 \quad (k = 1, 3).$$

Together with $\epsilon_5 = -1$, both odd channels have $(\epsilon_2, \epsilon_5) = (-1, -1)$. The signs still cannot distinguish $k = 1$ from $k = 3$, nor the coordinate labels 5 from 7; they factor only through the common sign C_2 .

Guard. The result is tied to the frozen different-normalized additive character and inherited ring-class character convention. No statement about a differently normalized automorphic/newform epsilon factor is made without an explicit normalization adapter.

8 Stone/Fibonacci project payload gate

A renewed same-carrier search finds no new Stone project selector beyond the actual A1BE/F1 partial representation packet and the universal secondary mapper. The still-missing fields are the project differential d , project θ , primitive K_Z , project coupling C and residue/anchor provenance. The historical contract remains $\lambda_\Theta = \theta K_Z \bmod 24$ once those bytes are supplied.

For Fibonacci the minimal decisive source packet remains

$$(m, \mathcal{U}, \mathcal{I}(m), \mathcal{I}(m) \cap \mathcal{U} = \emptyset).$$

The source theorem identifies this exactly with a direct RLM/support witness, but current storage still contains no actual same-carrier instance. Therefore

$$\text{ABI-complete project packets} = 0, \quad \text{promoted project values} = 0,$$

and all 37 active project gates remain open.

9 Checkpoint and safe successor

II-JU appends

$$992 \text{ PASS}, \quad 84 \text{ ZERO}, \quad 293 \text{ SEPARATED}, \quad 259 \text{ REFUTED}, \quad 38 \text{ NT}$$

and orthogonally

$$502 \text{ PROVEN-MERGEABLE}, \quad 319 \text{ PROVEN-NOT-MERGEABLE}.$$

The deep verifier closes 11 evidence families with 409 132 runtime assertion events, 385 explicit witnesses and 1 043 072 coverage units. Observational continuity is `PASS_280`; historical exhaustive replay remains `NT_NOT_EXHAUSTIVE`.

II-JV – Cube27 Picard-Orbit Tensor Autoequivalence and Stable Green-Ring Shadow; Gauss-Tail Transfer-Operator Spectrum/Variance and Khinchin–Lyapunov Refinement; Codimension-3 Full D_4 Flow-Covariance/Orbit-Atlas Cocycle; Golden-Hodge Native Comparison-Cochain Solver and H^4 Ring Closure; PLR Conditional C_{11} versus C_5/C_7 Framing Descent and Golden 1 + 2 Selector Search; Pisano–Theta Local-to-Global Epsilon Normalization Adapter; plus Stone Secondary-Chain Source Mapper Completion and Fibonacci Same-Carrier RLM Witness Constructor toward First ABI-Complete Project Value.

References

- [1] Nico59000/Modular-recurrence-symetry-analyzer-, pinned at commit 9da771f5ea3fc09693d7d179042ab3bc2c02dd2c; connector snapshot and per-file Git SHAs are frozen in `source_recovery/github`.
- [2] D. B. A. Epstein and E. Vogt, *A Counterexample to the Periodic Orbit Conjecture in Codimension 3*, Ann. of Math. 108 (1978), 539–552. Hash-locked source in `source_recovery/codim3`.
- [3] D. Sullivan, *A Counterexample to the Periodic Orbit Conjecture*, Publ. Math. IHES 46 (1976), 5–14. Hash-locked source in `source_recovery/codim3`.
- [4] J. Tate, *Fourier Analysis in Number Fields and Hecke’s Zeta-Functions*, in Cassels–Frohlich, *Algebraic Number Theory*; used only for the ramified local epsilon/Gauss-sum normalization principle.
- [5] II-JQ–II-JT frozen bundles and II-ID/II-IK/II-HQ source contracts, hash-locked in `dependencies` and the retrieval provenance ledger.